Assignment # 1

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SP25-BCS-112

Submitted To:

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Course name/code:

Data Structures

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HTML String Validator

CODE...

Main..

1.
```
import java.util.Scanner;
public class main {
  public static void main(String[] args) {
  Scanner sc=new Scanner(System.in);
  Validator validator=new Validator();
  LinkedList list=new LinkedList();

    System.out.println("........................................");
    System.out.println(".............HTML Validator.............");
    System.out.println("........................................");
    while (true){
    System.out.println("\nMenu....");
    System.out.println("  1...Check Validation of an HTML String.");
    System.out.println("  2...View History.");
    System.out.println("  3...Exit.");
    System.out.println("\nEnter Choice: ");
    int choice= sc.nextInt();
    sc.nextLine();

    switch (choice){
      case 1:
        System.out.println("Enter the HTML String : ");
        String html=sc.nextLine();
        String result=validator.Validation(html);
        System.out.println("Result : "+result);
        list.InsertAtLast(html,result);
        break;
```

```
            case 2:
              System.out.println("------Validation History------");
              list.display();
              break;
            case 3:
              System.out.println("Exiting...");
              return;
            default:
              System.out.println("Invalid Choice...");
          }
        }
        }
      }
```

Node...

2.
```
public class Node {
  Node nextaddress;
  String html_string;
  String validation;

  public Node(String html_string, String validation){
    this.html_string=html_string;
    this.validation=validation;
  }

}
```

LinkedList(History)..

3.
```
public class LinkedList {
  Node head;
  int size=0;

  public void InsertAtLast(String html,String validation){
    Node node=new Node(html,validation);
    if(head==null){
      head=node;
      size++;
      return;
```

```
        }
        Node temp=head;
        while(temp.nextaddress!=null){
            temp=temp.nextaddress;
        }
        temp.nextaddress=node;
        size++;
    }
    public void display(){
        Node temp=head;
        int count=1;
        while (temp!=null){
            System.out.println("........."+count+++".........");
            System.out.println("Given String : "+temp.html_string);
            System.out.println("Validation : "+temp.validation);
            temp=temp.nextaddress;
        }
        System.out.println("List Finished...");
    }
}
```

Stack...

4.
```
public class stack {
    int top;
    String stack[];
    int size;

    public stack(int size){
    this.top=-1;
    this.size=size;
    this.stack=new String[size];
    }

    public void push(String input){
        if(top==size-1){
            String stack2[]=new String[size*2];
            int count=0;
            for (int i=0;i<stack.length;i++){
```

```
            stack2[count++]=stack[i];
          }
          stack=stack2;
          size=size*2;
        }
        stack[++top]=input;
      }

      public String pop(){
        if(top==-1){
          return null;
        }
        return stack[top--];
      }
      public Boolean isEmpty(){
        return top==-1;
      }
      public String peek(){
        return stack[top];
      }
    }
```

Validator Class…

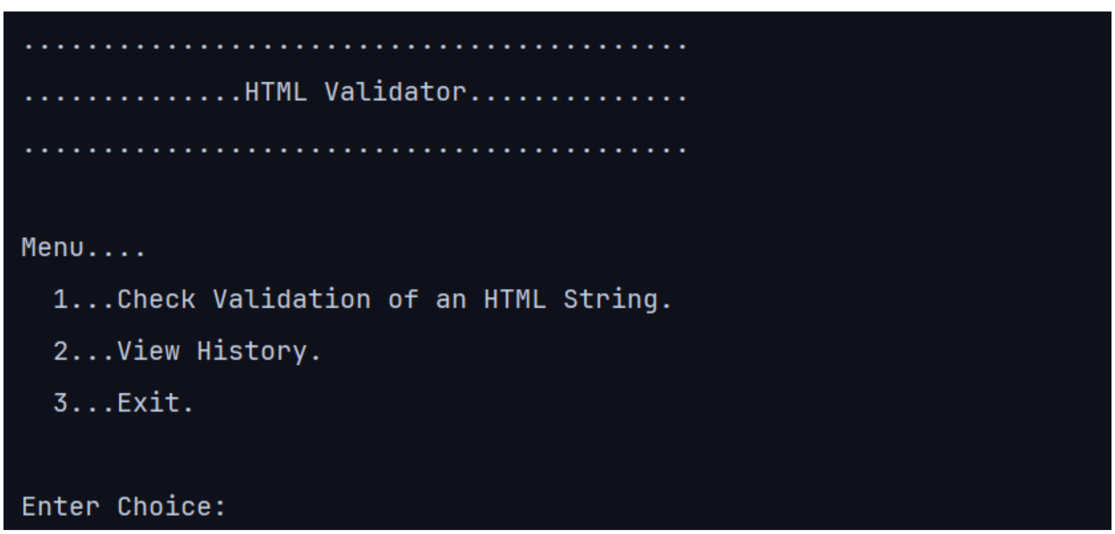
5. ```
public class Validator {

      public String Validation(String html_string){
        stack s=new stack(100);
        int i=0;
        while (i<html_string.length()){
          char ch=html_string.charAt(i);
          if(ch=='<') {
            int close = html_string.indexOf('>', i);
            if (close == -1) {
              return "Invalid HTML-String";
            }
            String opening_tag = html_string.substring(i + 1, close);
            i = close + 1;
            if (opening_tag.startsWith("/")) {
```
```

```
                String closingtag = opening_tag.substring(1);
                if (s.isEmpty() || !s.peek().equals(closingtag)) {
                  return "Invalid HTML-String";
                }
                s.pop();
              } else {
                s.push(opening_tag);
              }
            }
            else {
              i++;
            }
            }
          return s.isEmpty()? "Valid HTML-String":"Invalid HTML-String";
        }

      }
```

OUTPUT...



```
..........................................
..............HTML Validator..............
..........................................

Menu....
  1...Check Validation of an HTML String.
  2...View History.
  3...Exit.

Enter Choice:
```

```
..........................................
..............HTML Validator..............
..........................................

Menu....
  1...Check Validation of an HTML String.
  2...View History.
  3...Exit.

Enter Choice:
1
Enter the HTML String :
<html><body>MyName</body></html>
Result : Valid HTML-String
```

```
Menu....
  1...Check Validation of an HTML String.
  2...View History.
  3...Exit.

Enter Choice:
2
------Validation History------
..........1..........
Given String : <html><body>MyName</body></html>
Validation : Valid HTML-String
List Finished...
```